

VAE

1) Model design

↓

Train the model

↓

Use the model

- Data understanding
- Data generation

Maximum likelihood estimation (MLE)

$$\begin{aligned} \max_{\Theta} \quad & \sum_{n=1}^N \log p_{\Theta}(x_n) \\ = \quad & \sum_{n=1}^N \log \int p_{\Theta}(x_n, z_n) dz_n \\ = \quad & \sum_{n=1}^N \log \int dz_n \frac{p_{\Theta}(x_n, z_n)}{q_{\lambda}(z_n|x_n)} q_{\lambda}(z_n|x_n) dz_n \\ = \quad & \sum_{n=1}^N \log E_{q_{\lambda}(z_n|x_n)} \left[\frac{p_{\Theta}(x_n, z_n)}{q_{\lambda}(z_n|x_n)} \right] \end{aligned}$$

$$\begin{aligned} & \geq \sum_{n=1}^N E_{q_{\lambda}(z_n|x_n)} \left[\log \frac{p_{\Theta}(x_n, z_n)}{q_{\lambda}(z_n|x_n)} \right] \\ & = \sum_{n=1}^N \left(E_{q_{\lambda}(z_n|x_n)} \left[\log p_{\Theta}(x_n, z_n) \right] - E_{q_{\lambda}(z_n|x_n)} \left[\log q_{\lambda}(z_n|x_n) \right] \right) \\ & = L(\Theta, \lambda) \end{aligned}$$

Expectation-maximization algorithm

While not converged

E-step: fix Θ , optimize λ

M-step: fix λ , optimize Θ

$\max_{\Theta, \lambda} L(\Theta, \lambda)$

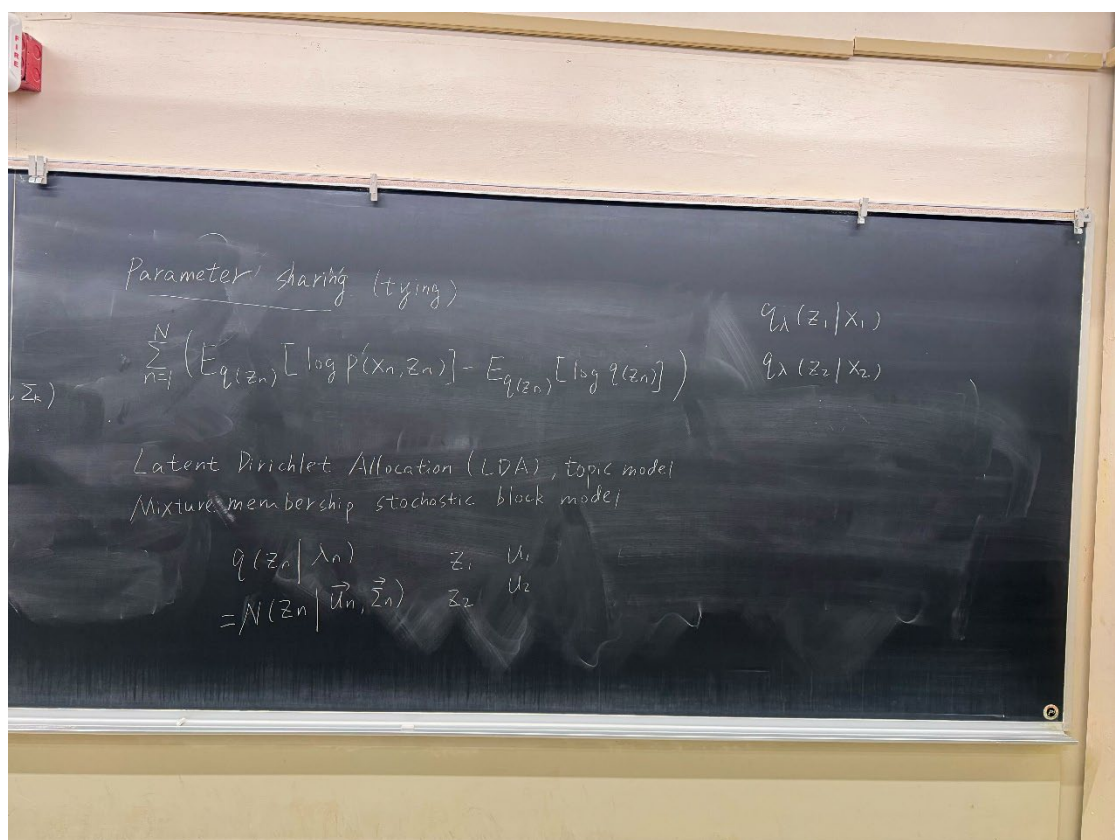
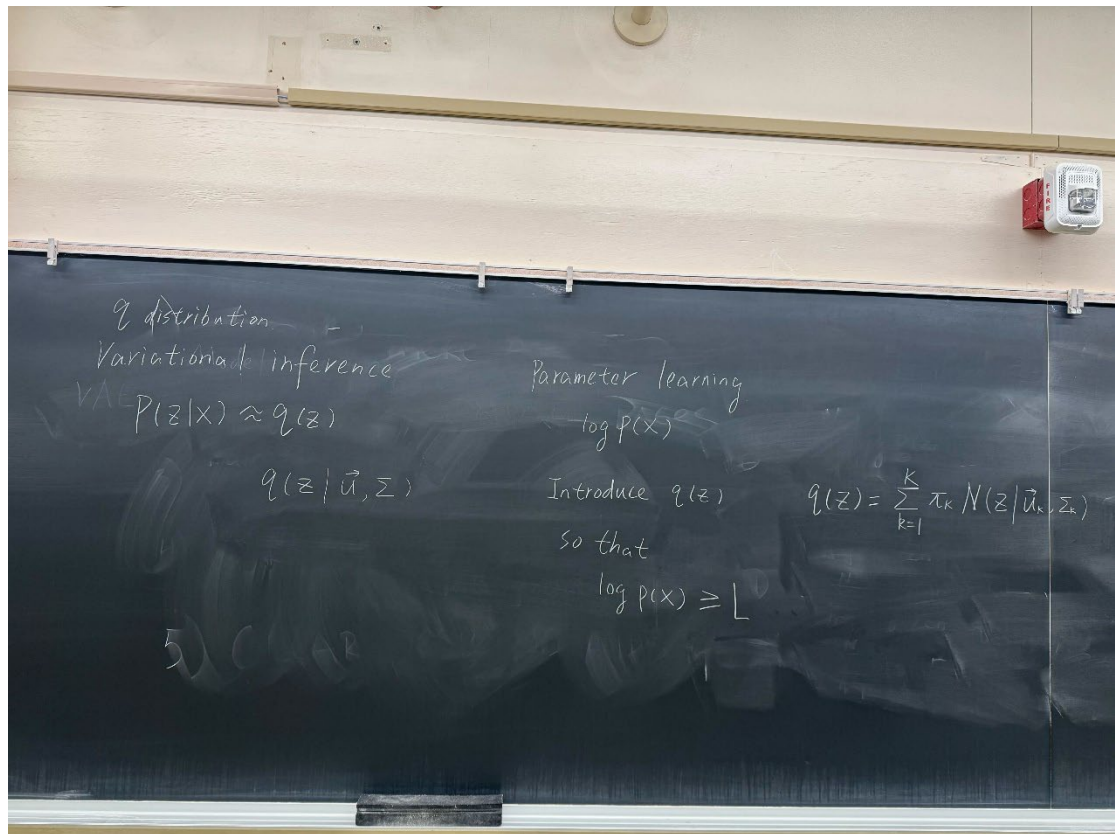
Diagram:

```

graph TD
    Xn[Xn] --> Qn[Qn]
    Qn --> Zn[Zn]
    Qn --> Pn[Pn]
    style Qn fill:#fff,stroke:#000,stroke-width:2px
    style Zn fill:#fff,stroke:#000,stroke-width:2px
    style Pn fill:#fff,stroke:#000,stroke-width:2px
  
```

$N(z_n | \hat{\alpha}(x_n), \text{diag}(\hat{\beta}(x_n)))$

$= q_{\lambda}(z_n|x_n)$



After parameter learning,
do we need to conduct
Optimization in variational inference?

$X_1, X_2, \dots, X_N$

$q_\lambda(z_i | X_i)$

$q_\lambda(z_N | X_N)$

X_{N+1}, X_{N+2}

$q(z_{N+1}) \quad q(z_{N+2})$

$\min KL(q(z_{N+1}) || P(z_{N+1} | X_{N+1}))$

Traditional

$q(z_{N+1} | \lambda_{N+1})$

VAE

$q_\lambda(z_{N+1} | X_{N+1})$

Turing test

tester

machine

human

human

DGM → synthetic image

→ real image
real images

Generative Adversarial Network (GAN)

training
real
images

→ real image

$z \sim N(0, I)$

Generator

$P(x|z)$

→ synthetic image

image → Discriminator → binary label

1: synthetic image
0: real image

